Emotional AI (Emotion Detection & Emotional TTS)

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What is it?

Input AI

Product: Input AI - Emotion Detection - Product Playbook - Cerence Drive - product - Cerence Confluence

Output AI

Product: (READY) Output AI - Product Playbook

Sales Training: 2024-01-30: Update on Neural TTS (internal)

Web Demo: Cerence Neural TTS 23.09.0 (TTS R&D Web Demo) - Internal

Preset Styles information: Emotional Neural TTS - Preset Styles

Release information: Cerence Neural TTS

(**Internal presentation with usecases**: Cerence bundle offer_Humanized Avatar_Exploratory conditions.pptx (sharepoint.com); restricted)

General info

Detection based on every speech in the car when listening e.g. for WuW, so no need to speak a command, also proactive dialogs could be triggered using the emotion.

Dialog can react on emotions:

1. Configure emotional style of TTS

2. Use different prompts depending on emotion
3. Modify dialog flow
4. Trigger proactive dialog

Usecase examples

Examples (by Vania, out of presentation above):

Description
Detecting Stress: The system can recognize when a driver is stressed and could proactively recommend solutions (e.g. for EV charging) or steps to create a more calming environment, for example, by adjusting the interior lighting or playing soothing music.
Preventing Accidents: Emotion detection can identify aggressive driving behavior linked to road rage, enabling the vehicle to intervene to ensure safety.
Improving Human-Machine Interaction: Emotion-aware systems can enhance the user experience by personalizing the in-car environment based on the driver's mood.
Preventing Fatigue Driving: Real-time emotion recognition can identify signs of driver fatigue, enabling the vehicle to take proactive measures such as alerting the driver or suggesting a break.

User Utterance	Detected Emotion	VA Utterance	TTS Emotion
		I looked at your calendar and I noticed that you have a meeting at the Cerence office in Burlington. Should I start navigation?	**Cheerful**
"Yes"	Approval	Your ETA is 10:20. Unfortunately you will arrive late for your appointment.	**Empathetic**

“I hate to be late! Can you find another route?”	Anger	I'm sorry to hear that. I couldn't find a better route, but would you like me to send a message to the meeting organizer to notify them that you'll be late?	**Sad** **Empathetic**
“Yes please”	Neutral	I sent them a message with your ETA.	**Neutral**

User Utterance	Detected Emotion	VA Utterance	TTS Emotion
		There's a risk of traffic jams due to two earlier accidents.	**Sad**
“Again? This is so frustrating”	Anger	Don’t worry, you should still make it on time. Would you like to take a break at your favorite coffee place?	**Empathetic**
“That sounds lovely!”	Love	Would you like to add it as a stop, or set it as your new destination?	**Cheerful**
“Go directly”	Neutral	Redirecting to "Best Coffee in Town". <pause 400> Should I also change the ambient lights to a relaxing color for you?	**Neutral**
“Sounds great!”	Admiration	Turning the ambient lights to blue. Enjoy!	**Cheerful**

User Utterance	Detected Emotion	VA Utterance	TTS Emotion
		You seem to be stressed out. I don’t like traffic jams either. Should I find a route that avoids the traffic?	**Empathetic**

“Sounds amazing”	Excitement	Ah I'm sorry. It looks like there isn't an alternative route available.	**Empathetic**
“This is really annoying!”	Annoyance	I'm sorry I was unable to help.	**Empathetic**

User Utterance	Detected Emotion	VA Utterance	TTS Emotion
		You seem to be stressed out. I don’t like traffic jams either. Should I find a route that avoids the traffic?	**Empathetic**
“I am not interested in this kind of notifications”	Disapproval	Alright, I'll take note and update your preferences on proactive messages.	**Neutral**

Other info

Mark Kane:

Here is some initial information.

1. Our Emotion detection is based on stereotypical emotions. If users' emotions are weak or not clear then the model will not give clear results.
2. Collecting happy data: if you're happy and smiling while you're talking, the model will have a better chance detecting that you're happy i.e. you need to be genuinely happy.
3. Collecting angry data: people don't like showing their true angry side, as such, pseudo angry attempts might not be detected...let your anger out.

We have tested the solution on many datasets, however these datasets were often created by voice actors who know how to communicate emotion on demand. This can be a little be trickier for you and I. I recommend to listen to other stereotypical happy/angry content to help you mimic that emotion.

Regarding stereotypical content, just think about obvious speech emotion observed on TV e.g. how someone is obviously angry or happy. If the spoken emotion is subtle or nuanced it may not be detected. Stereotypical content for happy would be genuinely smiling when you're speaking. For anger its not just not smiling (as that could be neutral), you need to mean that anger. Make sure you're not confusing upset, irritated, annoyed, or frustrated with anger.

- Can you provide me some examples of stereotypical contents?

Please see attached some recording I made for you.

angry1.wav angry2.wav angry3.wav happy1.wav happy2.wav

- Can you tell me in which languages Emotion AI is used?

The v0.91 solution has been shown to work explicitly on English and German. We've also shown that this solution is somewhat language independent by testing it on languages not included within the training set e.g. south American language (Quechua), a European language (French), and

a south Asian language (Bangla). We continue to work on improving this model explicitly against other languages.

- Can you give me criteria for how will the model detected the emotions?

The model is trained on a range of languages which lends itself to being somewhat language independent. The model expects a speech utterance between 1 and 4 seconds in length with leading and trailing silences removed.

Emotional Neural TTS - Preset Styles

Style name	Description	Example (US English Zoe) / Taken from Sales Training 2024 slides)
Cheerful	Expresses a positive and happy tone	Low: cheerful_low.wav Medium: cheerful_medium.wav High: cheerful_high.wav
Empathetic	Expresses a sense of caring and understanding	Low: empathetic_low.wav Medium: empathetic_medium.wav High: empathetic_high.wav
Sad	Expresses a sorrowful tone	Low: sad_low.wav

		Medium: sad_medium.wav High: sad_high.wav
Serious	Expresses a strict and commanding tone. Speaker often sounds stiffer and much less relaxed with firm cadence	Low: serious_low.wav Medium: serious_medium.wav High: serious_high.wav
Apprehensive	Anxious or fearful that something bad or unpleasant will happen.	Low: apprehensive_low.wav Medium: apprehensive_medium.wav High: apprehensive_high.wav
Conversational	This is intended as general purpose style, potentially applicable to any domain, and to both short/long sentences	conversational.wav

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Effects of Emotional TTS to the User Experience (User Research Overview)

Trust & Acceptance

- Users prefer **emotional voices** that sound **positive or neutral**
- greater trustworthiness & attractiveness (Pias et al., 2024)
- higher acceptance (Kim & Yoo, 2024)
- especially in emotionally negative contexts (Ma et al., 2025)
- Users prefer **emotional voices** that **mirror their own mood (= emotional congruence)** (Zhao et al., 2024; Zhang et al., 2025), associated with increased trust in the system & greater acceptance of the interaction.
- Users do **not prefer voices** that sound **exaggerated**. Mismatched tones can reduce comfort & trust.

Driving Performance & Safety

- TTS styles **adapted to the user's emotional state** improve **takeover performance, situational awareness, & driving behavior** (Nass et al., 2005; Zhao et al., 2024; Braun et al., 2022)

- Especially in autonomous driving modes, an emotional assistant helps regulate negative emotions & shorten reaction times (Zhao et al., 2024)
- **Speech output** modeled with **anger** can increase **driver alertness levels & reduce distracted driving** (Hsieh et al., 2010; Braun et al., 2022).

🌐 **Cultural Differences**

- **Positive** emotional **expressions** (cheerful, calm) are **preferred across cultures** (Kim & Yoo, 2024)
- **Negative** emotional **expressions** (sad, angry) are met **with low acceptance**, particularly in **Western** cultures (Kim & Yoo, 2024)

➡️ According to the Media Equation Theory—which suggests that people tend to treat systems like they would other humans—an emotionally responsive voice assistant can create positive effects by appearing more "human." It can enhance trust and likability, improve concentration, and increase safety. Research recommends using a natural, emotionally balanced tone that is context-sensitive and adaptively applied.

UX Design Recommendations

❗ **Apply Emotional Styles to prompts where context is clear**

- For navigation & in-car functionality, use conversational style for clear, warm, and non-intrusive prompts that support focus and comfort. Serious for safety-critical prompts, cheerful for entertainment, empathic for support. Avoid sad and apprehensive styles, since they might be less accepted by users.

❌ **Avoid emotional extremes**

- Voice Assistant Personas should maintain a natural and emotionally balanced tone - neither overly energetic nor too subdued. Users perceive such voices as more trustworthy, likable, and less distracting, especially in driving contexts where emotional extremes can reduce comfort and impair focus.

👩‍💻 **Support the assistant's persona (persona consistency)**

- The voice must match the desired persona (e.g., classy, confident, mindful). The emotional style reinforces the brand identity.

😉 **Align with the driver's emotional state when possible**

- Adjusting the assistant's voice style to reflect or balance the user's emotional state is a key factor in interaction quality. Emotional congruence fosters trust, enhances the sense of being understood, increases engagement with voice UIs, and helps drivers maintain greater focus on the driving task.

🗣️ **Orthography & Prosody**

- Ensure correct spelling, punctuation and setting of pauses to ensure natural rhythm. Well-structured prosody enhances intelligibility and reduces driver distraction.

🔤 **Text design**

- Consider semantics and emotional nuance of prompts. The voice should reinforce—not contradict—the intended message.

🧑‍🎨 **Variation & Adaptability**

- Continuously experiment with text formulations, voice styles, and pause placements to optimize clarity, naturalness, and emotional resonance.

⏸️ **Consider natural Pause Placement**

- Insert pauses only where they naturally occur (sentence breaks, semantic units) to improve comprehension without disrupting the flow.

📋 **Validate the UX regularly**

- Test emotional styles in real driving scenarios from a UX perspective to ensure they do not feel disruptive or artificial.
- Use A/B tests, Wizard-of-Oz experiments, and user interviews for optimization.

See also

Proposal for Tests

https://cerence.atlassian.net/wiki/download/attachments/1519977254/Emotion_Detection_Tests et_PS_Requirements_Measurement.docx?api=v2 (doc to support PS testing, by Daniel Kindermann).

Demo MBUX Virtual Assistant at CES2024

https://www.linkedin.com/feed/update/urn:li:activity:7155517089057820673?updateEntityUrn=urn%3Ali%3Afs_feedUpdate%3A%28V2%2Curn%3Ali%3Aactivity%3A7155517089057820673%29

EMMI (Empathic Human-Machine Interaction) Project

https://www.emmi-projekt.de/newsblog.html

Effects of emotional TTS to the User Experience (high level research overview) - From Mercedes-Benz TTS/UX Workshop September 2025

Effects of emotional TTS to the User Experience.pptx

Related External Research

Title, Citation	Abstract
At Your Service: Designing Voice Assistant Personalities to Improve Automotive User Interfaces. **Braun, M.**, Mainz, A., Chadowitz, R., Pfleging, B., & Alt, F. (2019). At Your Service: Designing Voice Assistant Personalities to Improve Automotive User Interfaces. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–11. *Full text: here*	This paper investigates personalized voice characters for incar speech interfaces. In particular, we report on how we designed different personalities for voice assistants and compared them in a real world driving study. Voice assistants have become important for a wide range of use cases, yet current interfaces are using the same style of auditory response in every situation, despite varying user needs and personalities. To close this gap, we designed four assistant personalities (Friend, Admirer, Aunt, and Butler) and compared them to a baseline (Default) in a between-subject study in real traffic conditions. Our **results show higher likability and trust for assistants that correctly match the user's personality while we observed lower likability, trust, satisfaction, and usefulness for incorrectly matched personalities, each in comparison with the Default character.** We discuss design aspects for voice assistants in different automotive use cases.
Affective Automotive User Interfaces—	Affective technology offers exciting opportunities to improve road safety by catering to human emotions. Modern car interiors enable

Reviewing the State of Emotion Regulation in the Car **Braun, M.**, Weber, F., & Alt, F. (2022). Affective Automotive User Interfaces—Reviewing the State of Emotion Regulation in the Car. *ACM Computing Surveys*, *54*(7), 1–26. *Full text: here*	the contactless detection of user states, paving the way for a systematic promotion of safe driver behavior through emotion regulation. We review the current literature regarding the impact of emotions on driver behavior and analyze the state of emotion regulation approaches in the car. We summarize challenges for affective interaction in form of cultural aspects, technological hurdles and methodological considerations, as well as opportunities to improve road safety by reinstating drivers into an emotionally balanced state. The purpose of this review is to outline the community's combined knowledge for interested researchers, to provide a focussed introduction for practitioners and to identify future directions for affective interaction in the car.
Effect of emotional speech tone on driving from lab to road: fMRI and ERP studies. **Hsieh, L.**, Seaman, S., & Young, R. (2010). Effect of emotional speech tone on driving from lab to road: fMRI and ERP studies. *Proceedings of the 2nd International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 22–28. *Full text: here*	Evoked Response Potential (ERP) and functional Magnetic Resonance Imaging (fMRI) recordings in this study shed light on underlying neural mechanisms for higher cognitive processes and attention allocation during multitasking of cell phone conversations and driving. Behavioral results indicated that hands-free cellular phone conversations caused statistically significant but small reaction time effects for visual event detection during simulated and on-road driving. The validated Static Load driving paradigm gives rise to high correlations of red light reaction times between lab and on-road. Both ERP and fMRI findings suggested that cognitive distractions are correlated with increased cognitive load and attentional distribution. **The novel contribution of this ERP and fMRI study is that adding an angry emotional valence to the speech increased the alertness level, resulting in reduced driver distraction, likely via increases in right frontoparietal networks and dampened or desynchronized left frontal activity.**
Acceptability Study of Emotional Expressions of In-Vehicle Virtual Assistant.	We analyzed the differences in drivers' preferences for emotional expressions from an in-vehicle virtual assistant based on data collected from 823 respondents in five countries categorized into three different cultural settings: collectivism (Korea, China and Brazil), individualism (USA) and a mixture of both (Russia). To quantify the degree of acceptance, we used a 5-point Likert scale

Kim, Y., & Yoo, H. (2024). Acceptability Study of Emotional Expressions of In-Vehicle Virtual Assistant. *Interacting with Computers*, *36*(3), 188–197. Acceptability Study of Emotional Expressions of In-Vehicle Virtual Assistant	(1 = strongly disagree; 5 = strongly agree) on four emotion types categorized based on Thayer's two-dimensional emotion model. **Analysis revealed that respondents in all five countries preferred emotion type 1 (pleased, happy, excited) and emotion type 4 (calm, peaceful, relaxed), that is, positive emotions under Thayer's two-dimensional emotion model expressions, and they showed low acceptance for emotion type 2 (nervous, angry, annoying) and emotion type 3 (sad, bored, sleepy), negative emotional expressions. In a cross-cultural comparative analysis, Brazil and China ranked highest in both positive emotion types (1 and 4), and Korea ranked highest in negative emotion types.**
Emotion-Aware Voice Assistants: Design, Implementation, and Preliminary Insights. **Ma, Y.**, Zhang, Y., Bachinski, M., & Fjeld, M. (2023). Emotion-Aware Voice Assistants: Design, Implementation, and Preliminary Insights. *Proceedings of the Eleventh International Symposium of Chinese CHI*, 527–532. https://doi.org/10.1145/3629606.3629665	The increasing prevalence of voice assistants in daily life has transformed human-technology interaction. However, current voice assistants fall short in their ability to comprehend users' emotional states through speech and respond empathetically. This limitation leads to suboptimal user experiences and missed opportunities for personalization. To address this gap, this paper presents a groundbreaking prototype of a future voice assistant equipped with the capability to recognize and respond to users' emotional states. The paper outlines the architecture of a speech-emotion recognition system that utilizes a deep neural network incorporating a sparse attention mechanism, yielding promising results. Furthermore, the paper embarks on an initial exploration of strategies for emotionbased responses, drawing inspiration from human techniques for managing unpleasant emotions. A user study is conducted, exposing participants to emotions such as anger, sadness, and fear, while engaging the voice assistant to elicit happiness. **Preliminary findings suggest a tendency toward neutral emotional responses**, providing valuable insights into the potential of speech feature analysis as an alternative approach for identifying and responding to emotions.
Advancing User-Voice Interaction: Exploring Emotion-Aware Voice Assistants Through a	As voice assistants (VAs) become increasingly integrated into daily life, the need for emotion-aware systems that can recognize and respond appropriately to user emotions has grown. While significant progress has been made in speech emotion recognition (SER) and sentiment analysis, effectively addressing user emotions-particularly

Role-Swapping Approach **Ma, Y.**, Zhang, Y., Fu, D., Portales, S. Z., Kragic, D., & Fjeld, M. (2025). *Advancing User-Voice Interaction: Exploring Emotion-Aware Voice Assistants Through a Role-Swapping Approach* (Version 1). arXiv. Advancing User-Voice Interaction: Exploring Emotion-Aware Voice...	negative ones-remains a challenge. This study explores human emotional response strategies in VA interactions using a role-swapping approach, where participants regulate AI emotions rather than receiving pre-programmed responses. Through speech feature analysis and natural language processing (NLP), we examined acoustic and linguistic patterns across various emotional scenarios. **Results show that participants favor neutral or positive emotional responses when engaging with negative emotional cues, highlighting a natural tendency toward emotional regulation and de-escalation.** Key acoustic indicators such as root mean square (RMS), zero-crossing rate (ZCR), and jitter were identified as sensitive to emotional states, while sentiment polarity and lexical diversity (TTR) distinguished between positive and negative responses. These findings provide valuable insights for developing adaptive, context-aware VAs capable of delivering empathetic, culturally sensitive, and user-aligned responses. By understanding how humans naturally regulate emotions in AI interactions, this research contributes to the design of more intuitive and emotionally intelligent voice assistants, enhancing user trust and engagement in human-AI interactions.
How do voice acoustics affect the perceived trustworthiness of a speaker? A systematic review. **Maltezou-Papastylianou, C.**, Scherer, R., & Paulmann, S. (2025). How do voice acoustics affect the perceived trustworthiness of a speaker? A systematic review. *Frontiers in Psychology*, *16*, 1495456. Frontiers | How do voic	Trust is a multidimensional and dynamic social and cognitive construct, considered the glue of society. Gauging someone's perceived trustworthiness is essential for forming and maintaining healthy relationships across various domains. Humans have become adept at inferring such traits from speech for survival and sustainability. This skill has extended to the technological space, giving rise to humanlike voice technologies. The inclination to assign personality traits to these technologies suggests that machines may be processed along similar social and vocal dimensions as human voices. Given the increasing prevalence of voice technology in everyday tasks, this systematic review examines the factors in the psychology of voice acoustics that influence listeners' trustworthiness perception of speakers, be they human or machine. Overall, this systematic review has revealed that **voice acoustics impact perceptions of trustworthiness in both humans and machines. Specifically, combining multiple acoustic features through multivariate methods enhances interpretability and yields**

e acoustics affect the per ceived trustworthiness of a speaker? A systematic review	**more balanced findings compared to univariate approaches.** Focusing solely on isolated features like pitch often yields inconclusive results when viewed collectively across studies without considering other factors. **Crucially, situational, or contextual factors should be utilised for enhanced interpretation as they tend to offer more balanced findings across studies.** Moreover, this review has highlighted the significance of cross-examining speaker-listener demographic diversity, such as ethnicity and age groups; yet, the scarcity of such efforts accentuates the need for increased attention in this area. Lastly, future work should involve listeners' own trust predispositions and personality traits with ratings of trustworthiness perceptions.
Improving automotive safety by pairing driver emotion and car voice emotion. **Nass, C.**, Jonsson, I.-M., Harris, H., Reaves, B., Endo, J., Brave, S., & Takayama, L. (2005). Improving automotive safety by pairing driver emotion and car voice emotion. *CHI '05 Extended Abstracts on Human Factors in Computing Systems*, 1973–1976. https://doi.org/10.1145/1056808.1057070	This study examines whether characteristics of a car voice can affect driver performance and affect. In a 2 (driver emotion: happy or upset) x 2 (car voice emotion: energetic vs. subdued) experimental study, participants (N=40) had emotion induced through watching one of two sets of 5-minute video clips. Participants then spent 20 minutes in a driving simulator where a voice in the car spoke 36 questions (e.g., "How do you think that the car is performing?") and comments ("My favorite part of this drive is the lighthouse.") in either an energetic or subdued voice. Participants were invited to interact with the car voice. **When user emotion matched car voice emotion (happy/energetic and upset/subdued), drivers had fewer accidents, attended more to the road (actual and perceived), and spoke more to the car.** Implications for car design and voice user interface design are discussed.
Building Trust Through Voice: How Vocal Tone Impacts User Perception of Attractiveness of Voice Assistants	Voice Assistants (VAs) are popular for simple tasks, but users are often hesitant to use them for complex activities like online shopping. We explored whether the vocal characteristics like the VA's vocal tone, can make VAs perceived as more attractive and trustworthy to users for complex tasks. **Our findings show that the tone of the VA voice significantly impacts its perceived attractiveness and**

Pias, S. B. H., Freel, A., Huang, R., Williamson, D., Kim, M., & Kapadia, A. (2024). *Building Trust Through Voice: How Vocal Tone Impacts User Perception of Attractiveness of Voice Assistants* (No. arXiv:2409.18941). arXiv. Building Trust Through Voice: How Vocal Tone Impacts User...	**trustworthiness. Participants in our experiment were more likely to be attracted to VAs with positive or neutral tones and ultimately trusted the VAs they found more attractive.** We conclude that VA's perceived trustworthiness can be enhanced through thoughtful voice design, incorporating a variety of vocal tones.
Voice Similarity and its Impact on Cognitive and Affective Trust in Automated Vehicles. **Zhang, Q.,** Yang, X. J., & Robert, L. P. (2025). Voice Similarity and its Impact on Cognitive and Affective Trust in Automated Vehicles. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 10711813251364804. Voice Similarity and its Impact on Cognitive and Affective Trust in Automated Vehicles - Qiaoning Zhang, X. Jessie Yang, Lionel P. Robert, 2025	Building user trust is critical for the widespread adoption of automated vehicles (AVs) as they become more integrated into our daily lives. This study explores how the voice used by AVs can influence two types of trust: cognitive trust (belief in the AV's competence and reliability) and affective trust (emotional connection with the AV). Drawing from similarity-attraction theory, the research investigates whether users are more likely to trust AVs whose voices match their age and gender. In an online study involving over 300 U.S. drivers, participants experienced AV explanations delivered in voices that aligned with or differed from their demographic characteristics. **The results revealed that users reported significantly higher cognitive and affective trust when the AV voice matched their own. Gender similarity strongly impacted both types of trust, while age similarity mainly affected affective trust.** These findings highlight the power of personalized voice design in making AVs feel more relatable and trustworthy. This research offers valuable insights for designers and developers aiming to enhance human-AV interaction through more socially attuned and emotionally resonant communication strategies.
Emotion-matched in-vehicle assistants: An	In conditional autonomous driving, drivers are prone to experiencing incidental emotions during non-driving tasks, which are unrelated to

exploration in regulating drivers' incidental emotions and enhancing takeover performance and situational awareness. **Zhao, Y.**, Jiang, K., Yu, Z., & Huang, Z. (2024). Emotion-matched in-vehicle assistants: An exploration in regulating drivers' incidental emotions and enhancing takeover performance and situational awareness. *Cognition, Technology & Work*, *26*(2), 225–245. https://doi.org/10.1007/s10111-024-00756-8	driving decisions but may have negative effects on drivers' takeover performance. Voice assistants have been shown to regulate incidental emotions. Therefore, this study endowed the in-vehicle assistant with emotional voice capability, matching the emotional state of the driver in conditional autonomous driving, in order to explore the effect of regulating incidental emotions. Forty participants were enrolled in a driving simulation experiment. During the autonomous driving process, participants were induced with two incidental emotions (happy or sad) and underwent experiments combining two emotional states of the driver and two emotional states of the in-vehicle assistant. The purpose was to examine the participants' takeover performance in autonomous driving, eye-tracking data, situation awareness, and to reveal the preferences of participants in different groups regarding the in-vehicle emotional voice assistant and their trust in the automated system. The results showed that **in-vehicle emotional voice assistants that matched drivers' incidental emotions exhibited better effects on emotional regulation, as evidenced by shorter takeover response time, smaller standard deviation of lane position, smaller maximum lateral acceleration, and better situation awareness capabilities. Drivers also prefer in-vehicle emotional voice assistants that have similar incidental emotions to theirs.** Interestingly, the regression model revealed that **drivers' incidental emotions and the in-vehicle emotional voice assistant affected the takeover response time in the takeover performance indices**, which was mediated by situation awareness. The results of this study are expected to provide some inspiration for the design of current in-vehicle voice assistants.